Fast Convergence Robust Kalman Filter Localization of Outliers Based on Cloud Model

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***Abstract*—Fusion localization based on Kalman filter is a commonly used method, especially in narrow and enclosed underground environments. The poor performance of sensors such as LiDAR, GPS and visual odometry makes SLAM based positioning methods unsuitable, and it is necessary to improve Kalman filtering. However, traditional Kalman filtering frameworks cannot distinguish between measurement outliers and state outliers caused by model uncertainty. Therefore, this article uses cloud models to better distinguish them and embeds an adaptive method to obtain a sliding window variational robust Kalman filter based on Student's t-noise modeling. Compared with the commonly used Extend Kalman filter algorithm, the improved algorithm proposed in this article reduces the maximum position error by 20.69% and the average position error by 44.37% when the probability of outliers is 0.1; When the probability of outliers increases to 0.3, the maximum position error decreases by 30.38%, and the average position error decreases by 51.45%. In addition, in terms of convergence speed, due to the introduction of the concept of cloud model in this article, the convergence speed can reach 20 times that of existing Kalman filter methods based on Student's t-noise distribution models. The effectiveness of the algorithm was demonstrated through simulation experiments and real machine positioning experiments.**

***Index Terms*—Cloud models, extend Kalman filter, outlier, student's t-noise modeling.**

I. Introduction

ACCURATE positioning capabilities are paramount for robotic systems performing essential inspection and upkeep tasks within complex subterranean infrastructure. The lack of reliable positional data during such operations is a major contributor to construction inefficiencies, including delays and budget escalations, ultimately hindering urban development and management effectiveness [1]. While robotic deployment offers significant advantages, the operational reality within cramped, moisture-laden underground passages exposes critical limitations in current localization methodologies [2]: inherent global positioning system (GPS) denial, compromised LiDAR performance due to occluding deposits, and unreliable barometric sensing under variable atmospheric conditions.

These harsh operating conditions make state estimation based on multisensor fusion the core of positioning systems—it must simultaneously compensate for the non-Gaussian characteristics of sensor noise and the uncertainty of motion models. Although the extended Kalman filter (EKF) is widely used due to its computational efficiency, its inherent Gaussian assumption faces a dual risk of failure in underground pipeline scenarios: on the one hand, the highly nonlinear dynamics of robot pipe wall contact make state prediction susceptible to model simplification errors, resulting in state outliers; On the other hand, sudden local turbulence or high-frequency vibrations during cutting operations can cause measurement outliers. For example, when a robot passes through a sediment accumulation area, the non-Gaussian state deviation caused by the slippage of the wheel odometer is coupled with the exponential weighting characteristics of the traditional EKF, which can cause the positioning error to accumulate over 1.5 m within 10 s. This error propagation mechanism seriously threatens the reliability of localization and urgently requires the development of robust estimation frameworks with outlier immunity.

Significant research efforts have been devoted to enhancing Kalman filtering robustness against outliers [3], [4], [5], [6], [7]. Core methodologies include: adjusting the measurement model through Kalman gain optimization to minimize mean square estimation error [8], integrating neural networks for real-time correction in Extended Exponentially Weighted Kalman Filter implementations [9], and detecting anomalies via Maximum Correntropy Criterion combined with Local Outlier Factor analysis [10]. Advanced strategies further employ M-estimation feedback mechanisms to reduce outlier misclassification [11], [12], or leverage antioutlier graph frequency domain filtering for signal smoothness constraints [13]. Recent innovations focus on covariance estimation frameworks [14], [15], [16], [17], Masreliez-Martin filter adaptations [18], and probability-driven outlier detection integrated with Kalman

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filter [19], [20]. Within EKF-based frameworks, notable solutions involve: hyperparameter determination for latent outlier variables using expectation-maximization algorithms [21], sliding window-quadratic programming optimization for outlier suppression [22], and specialized robust Kalman smoothers addressing both propagating and nonpropagating outliers [23]. Noise model reconstruction approaches, particularly Student's t-distribution adaptations, demonstrate growing prominence exemplified by the multimodel adaptive estimation framework utilizing Extended Student's t Filters for dynamic degree-of-freedom selection [24], [25]. Complementary techniques span saturation function integration [26], graph representation/segmentation methods [27], and Huber cubic Kalman filter implementations [28], collectively expanding the toolkit for outlier-resilient state estimation.

It is well known that filtering methods based on the assumption of Gaussian distribution, such as the EKF, are highly dependent on the statistical accuracy of the model and noise parameters and are extremely susceptible to the influence of heavy-tailed noise and outliers. In severe cases, this susceptibility can lead to algorithm divergence. Existing filtering methods based on the Student's t-distribution ensure robustness in the presence of outliers; however, these algorithms tend to have relatively slow convergence rates and still require significant computational power. Building upon prior studies [29], [30], [31], [32], [33], this work develops an extended Kalman filtering framework that adopts a Student's t-distributed noise model, integrated with variational inference through a sliding-window mechanism for joint state estimation. A novel dual anomaly detection method is proposed to address potential outliers in both state prediction and measurement domains. The core innovation lies in leveraging cloud model theory to compute three temporal metrics—expectation, entropy, and hyper-entropy—across consecutive state-measurement pairs within observation windows. These metrics collectively determine a credibility degree at each timestep, with the maximum-credibility value selected as the optimal estimate. When current credibility falls below historical maxima, the filter autonomously adjusts the noise covariance matrix to dynamically mitigate outlier impacts, enabling real-time robustness adaptation.

The key innovations and contributions of this study are as follows.

1) This article introduces the concept of cloud model and uses the mathematical characteristics of expectation, entropy, hyperentropic, and cloud droplet certainty to evaluate the trend of data changes within a data window. The uncertainty is used to evaluate the probability of data being outliers at each moment and dynamically adjust the parameters of the noise covariance matrix.
2) Due to the tail characteristic of exponential decay, traditional Gaussian based models exhibit severe vulnerability to outliers. To address this fundamental limitation, our innovation lies in introducing a Student's t-distribution system model with adaptive scale parameters and combining it with the extended Kalman filter method in practical engineering applications, achieving a more robust localization method under outlier conditions.
3) Through simulation experiments and real machine platform verification, it has been shown that compared with existing robust methods, the proposed filter has higher tracking accuracy and consistency, as well as faster convergence speed, but with a greater computational burden.

The remainder of this article is organized as follows. Section II derives a new filtering framework and analyzes the mechanism of outlier suppression along with computational complexity and provides a detailed introduction to the outlier probability calculation method based on cloud models, establishing the foundation for subsequent work. Section III compares the proposed method with existing approaches through both localization simulations and real-world robotic localization experiments, verifying the algorithm's practicality in engineering applications. Finally, Section IV summarizes this article.

II. Proposed Method

The purpose of this method is to identify outliers within a sliding window and to increase the corresponding noise covariance matrix to suppress them. Fig. 1 illustrates the cloud model-based robust Kalman filter framework. The system collects raw sensor data (potentially containing environmental outliers) and separates it into state variables (true system state) and noisy measurements. Using cloud model theory, it extracts key features—expectation, entropy, and hyper-entropy-to characterize data centrality, dispersion, and uncertainty. Data certainty is computed to identify outliers, while a sliding window dynamically processes recent data. Within a variational Bayesian (VB) framework, fixed-point iteration jointly updates state variables, noise covariance matrices, and auxiliary random variables until convergence. The VB module employs a Student's t-noise model (enhanced outlier robustness versus Gaussian models) and dynamically adjusts noise covariance matrices based on outlier probabilities to suppress their impact. Finally, the system outputs estimated positions and variance, delivering reliable localization for robotic systems in challenging environments. This integrated framework ensures effective outlier handling and precise state estimation.

A. Noise Modeling

To address the impact of outliers on localization performance, article [29] reconstructed the heavy-tailed noise model into a Student's t model, introduced sliding windows and auxiliary random variables, and ultimately employed VB inference to jointly estimate state variables. Building on this work, the noise model is reformulated as a Student's t model, with the state transition distribution and measurement likelihood distribution at time $j \in [k-L+1, k]$ are as follows:

$$\begin{cases} p\left(\mathbf{x}_j \mid \mathbf{x}_{j-1}, \mathbf{Q}_k\right) = \mathrm{St}\left(\mathbf{x}_j; \mathbf{F}_j \mathbf{x}_{j-1}, \mathbf{Q}_k, \omega\right) \\ = \int \mathrm{N}\left(\mathbf{x}_j; \mathbf{F}_j \mathbf{x}_{j-1}, \mathbf{Q}_k/\xi_j\right) \mathrm{G}\left(\xi_j; \omega/2, \omega/2\right) d\xi_j \\ p\left(\mathbf{z}_j \mid \mathbf{x}_j, \mathbf{R}_k\right) = \mathrm{St}\left(\mathbf{z}_j; \mathbf{H}_j \mathbf{x}_j, \mathbf{R}_k, v\right) \\ = \int \mathrm{N}\left(\mathbf{z}_j; \mathbf{H}_j \mathbf{x}_j, \mathbf{R}_k/\lambda_j\right) \mathrm{G}\left(\lambda_j; v/2, v/2\right) d\lambda_j \end{cases} \tag{1}$$

among them, x_j and z_j represent state variables and measurement quantities, F_j and H_j represent state transition

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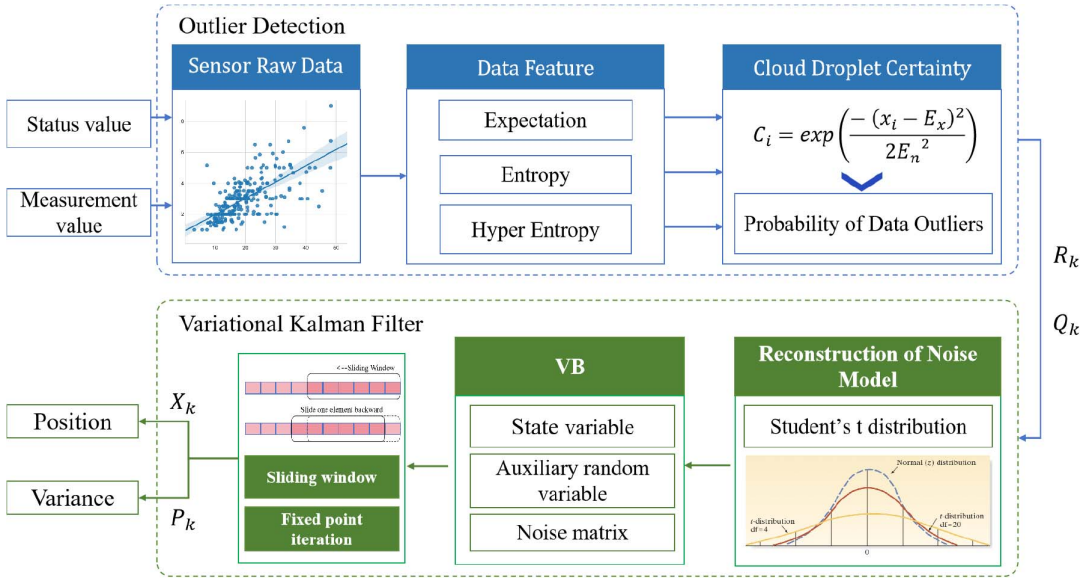


Fig. 1. System framework diagram.

matrices and observation matrices, Q_k and R_k represent process noise covariance matrices and measurement noise covariance matrices, ω and v represent the degrees of freedom of the distribution and introduce auxiliary random variables. To facilitate calculations, an auxiliary random variable is introduced based on the original article to express the Student's t-distribution in a Gaussian layered form.

According to the derivation results presented in article [29], the inverse Wishart (IW) distribution is chosen as the conjugate prior distribution for the normal distribution, which facilitates the application of VB inference process. Consequently, the following equation is derived as:

$$\begin{cases} p\left(\mathbf{Q}_k \mid \mathbf{z}_{1:k-L}\right) = \mathrm{IW}\left(\mathbf{Q}_k; \hat{y}_{k|k-L}, \hat{\mathbf{Y}}_{k|k-L}\right) \\ p\left(\mathbf{R}_k \mid \mathbf{z}_{1:k-L}\right) = \mathrm{IW}\left(\mathbf{R}_k; \hat{u}_{k|k-L}, \hat{\mathbf{U}}_{k|k-L}\right) \end{cases} \tag{2}$$

where $\hat{y}_{k|k-L}, \hat{u}_{k|k-L}, \hat{\mathbf{Y}}_{k|k-L}$, and $\hat{\mathbf{U}}_{k|k-L}$ respectively represent the prior parameters of Q_k and R_k. Due to the utilization of data from a specific time period, the forgetting factor ρ is introduced, and its variation is as follows:

$$\begin{cases} \hat{y}_{k|k-L} = \rho^L \hat{y}_{k-L|k-L}, & \hat{\mathbf{Y}}_{k|k-L} = \rho^L \hat{\mathbf{Y}}_{k-L|k-L} \\ \hat{u}_{k|k-L} = \rho^L \hat{u}_{k-L|k-L}, & \hat{\mathbf{U}}_{k|k-L} = \rho^L \hat{\mathbf{U}}_{k-L|k-L} \end{cases} \tag{3}$$

where $\hat{y}_{k-L|k-L}, \hat{u}_{k-L|k-L}, \hat{\mathbf{Y}}_{k-L|k-L}$, and $\hat{\mathbf{U}}_{k-L|k-L}$ are the posterior dofs and scale matrices of $\mathbf{Q}_{k-L}$ and $\mathbf{R}_{k-L}$.

Finally, set the initial value to the following form:

$$\begin{cases} p\left(\mathbf{Q}_0\right) = \mathrm{IW}\left(\mathbf{Q}_0; \hat{y}_{0|0}, \hat{\mathbf{Y}}_{0|0}\right), & \hat{\mathbf{Y}}_{0|0} = \hat{y}_{0|0}\boldsymbol{\Sigma}_w \\ p\left(\mathbf{R}_0\right) = \mathrm{IW}\left(\mathbf{R}_0; \hat{u}_{0|0}, \hat{\mathbf{U}}_{0|0}\right), & \hat{\mathbf{U}}_{0|0} = \hat{u}_{0|0}\boldsymbol{\Sigma}_v. \end{cases} \tag{4}$$

B. Variational Inference

To calculate the posterior probability distribution, this section employs the VB method to jointly estimate the state variables, the process noise covariance matrix, the measurement noise covariance matrix, and auxiliary random variables within a sliding window $[k-L+1, k]$. A window that is too long will consume excessive computational resources, leading to reduced real-time performance. Conversely, a window that is too short cannot capture complete outlier features. Relevant studies indicate that $L > 7$ is necessary to ensure sufficient historical information for identifying outliers. In this article, $L = 10$ is chosen. We define $\boldsymbol{\Theta}_k = \{\mathbf{x}_{k-L:k}, \mathbf{Q}_k, \mathbf{R}_k, \xi_{k-L+1:k}, \lambda_{k-L+1:k}\}$, because the estimated variables in VB methodology are considered pairwise independent, the joint posterior distribution can be approximated as [30]

$$\begin{aligned} p\left(\boldsymbol{\Theta}_k \mid \mathbf{z}_{1:k}\right) \approx\ & q\left(\mathbf{x}_{k-L:k}\right) q\left(\mathbf{Q}_k\right) \\ & q\left(\mathbf{R}_k\right) q\left(\xi_{k-L+1:k}\right) q\left(\lambda_{k-L+1:k}\right) \end{aligned} \tag{5}$$

according to Kullback-Leibler divergence calculation equation, every component $q\left(\boldsymbol{\theta}_k\right)$ is calculated as

$$\log q\left(\boldsymbol{\theta}_k\right) = \mathrm{E}_{(\boldsymbol{\Theta}_k - \boldsymbol{\theta}_k)}\left[\log p\left(\boldsymbol{\Theta}_k, \mathbf{z}_{1:k}\right)\right] + c_{\boldsymbol{\theta}_k} \tag{6}$$

among them, $\boldsymbol{\theta}_k$ is one of the elements in $\boldsymbol{\Theta}_k$, and $\mathrm{E}_x[\cdot]$ denotes the mathematical expectation of x. Consequently, the joint distribution in (6) is expressed as

$$\begin{aligned} p\left(\boldsymbol{\Theta}_k, \mathbf{z}_{1:k}\right) = & \prod_{j=k-L+1}^{k} \left[\sim \mathrm{N}\left(\mathbf{z}_j; \mathbf{H}_j \mathbf{x}_j, \mathbf{R}_k/\lambda_j\right)\right. \\ & \mathbf{G}\left(\lambda_j; \nu/2, \nu/2\right) \\ & \times \mathrm{N}\left(\mathbf{x}_j; \mathbf{F}_j \mathbf{x}_{j-1}, \mathbf{Q}_k/\xi_j\right) \mathrm{G}\left(\xi_j; \omega/2, \omega/2\right)] \\ & \times \mathrm{IW}\left(\mathbf{R}_k; \hat{u}_{k|k-L}, \hat{\mathbf{U}}_{k|k-L}\right) \end{aligned}$$

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$$\times \mathrm{IW}\left(\mathbf{Q}_k; \hat{y}_{k|k-L}, \hat{\mathbf{Y}}_{k|k-L}\right) \times \mathrm{N}\left(\mathbf{x}_{k-L}; \hat{\mathbf{x}}_{k-L|k-L}, \mathbf{P}_{k-L|k-L}\right). \tag{7}$$

Because each component in (5) is interdependent, the fixed-point iteration method is employed to calculate each $q(\boldsymbol{\theta}_k)$. To prevent redundancy, the detailed derivation process is referenced in article [29], and this section will only summarize the final results.

By setting $\boldsymbol{\theta}_k = \mathbf{x}_{k-L:k}$, we can derive the following equation:

$$\begin{cases} p\left(\mathbf{x}_{k-L} \mid \mathbf{z}_{1:k-L}\right) = \mathrm{N}\left(\mathbf{x}_{k-L}; \hat{\mathbf{x}}_{k-L|k-L}, \mathbf{P}_{k-L|k-L}\right) \\ \tilde{\mathbf{Q}}_{j|k}^{(i)} = \dfrac{\left\{\mathrm{E}^{(i)}\left[\mathbf{Q}_k^{-1}\right]\right\}^{-1}}{\mathrm{E}^{(i)}[\xi_j]}, \quad \tilde{\mathbf{R}}_{j|k}^{(i)} = \dfrac{\left\{\mathrm{E}^{(i)}\left[\mathbf{R}_k^{-1}\right]\right\}^{-1}}{\mathrm{E}^{(i)}[\lambda_j]} \end{cases} \tag{8}$$

where $j \in [k-L+1, k]$.

Similarly, by setting $\boldsymbol{\theta}_k = \mathbf{Q}_k$ and $\boldsymbol{\theta}_k = \mathbf{R}_k$, we can obtain

$$\begin{cases} q^{(i+1)}\left(\mathbf{Q}_k\right) = \mathrm{IW}\left(\mathbf{Q}_k; \hat{y}_{k|k}^{(i+1)}, \hat{\mathbf{Y}}_{k|k}^{(i+1)}\right) \\ q^{(i+1)}\left(\mathbf{R}_k\right) = \mathrm{IW}\left(\mathbf{R}_k; \hat{u}_{k|k}^{(i+1)}, \hat{\mathbf{U}}_{k|k}^{i+1)}\right) \end{cases} \tag{9}$$

the parameters can be updated in the following form:

$$\begin{cases} \hat{y}_{k|k}^{(i+1)} = \hat{y}_{k|k-L} + L \\ \hat{\mathbf{Y}}_{k|k}^{(i+1)} = \hat{\mathbf{Y}}_{k|k-L} + \sum_{j=k-L+1}^{k} \mathbf{A}_j^{(i+1)} \mathrm{E}^{(i)}[\xi_j] \\ \hat{u}_{k|k}^{(i+1)} = \hat{u}_{k|k-L} + L \\ \hat{\mathbf{U}}_{k|k}^{(i+1)} = \hat{\mathbf{U}}_{k|k-L} + \sum_{j=k-L+1}^{k} \mathbf{B}_j^{(i+1)} \mathrm{E}^{(i)}[\lambda_j] \end{cases} \tag{10}$$

where the auxiliary matrix $\mathbf{A}_j^{(i+1)}$ and $\mathbf{B}_j^{(i+1)}$ can be calculated using the following equation:

$$\begin{cases} \mathbf{A}_j^{(i+1)} = \mathrm{E}^{(i+1)}\left[\left(\mathbf{x}_j - \mathbf{F}_j \mathbf{x}_{j-1}\right)\left(\mathbf{x}_j - \mathbf{F}_j \mathbf{x}_{j-1}\right)^{\mathrm{T}}\right] \\ \mathbf{B}_j^{(i+1)} = \mathrm{E}^{(i+1)}\left[\left(\mathbf{z}_j - \mathbf{H}_j \mathbf{x}_j\right)\left(\mathbf{z}_j - \mathbf{H}_j \mathbf{x}_j\right)^{\mathrm{T}}\right]. \end{cases} \tag{11}$$

Let $\boldsymbol{\theta}_k = \xi_{k-L+1:k}$ and $\boldsymbol{\theta}_k = \lambda_{k-L+1:k}$, we can obtain

$$\begin{cases} q^{(i+1)}\left(\xi_j\right) = \mathrm{G}\left(\xi_j; \hat{a}_{j|k}^{(i+1)}, \hat{b}_{j|k}^{(i+1)}\right) \\ q^{(i+1)}\left(\lambda_j\right) = \mathrm{G}\left(\lambda_j; \hat{c}_{j|k}^{(i+1)}, \hat{d}_{j|k}^{(i+1)}\right). \end{cases} \tag{12}$$

The update method for each parameter is shown as

$$\begin{cases} \hat{a}_{j|k}^{(i+1)} = 0.5(\omega + n) \\ \hat{b}_{j|k}^{(i+1)} = 0.5\left(\omega + \mathrm{tr}\left(\mathbf{A}_j^{(i+1)} \mathrm{E}^{(i)}\left[\mathbf{Q}_k^{-1}\right]\right)\right) \\ \hat{c}_{j|k}^{(i+1)} = 0.5(v + m) \\ \hat{d}_{j|k}^{(i+1)} = 0.5\left(v + \mathrm{tr}\left(\mathbf{B}_j^{(i+1)} \mathrm{E}^{(i)}\left[\mathbf{R}_k^{-1}\right]\right)\right). \end{cases} \tag{13}$$

By combining (8)–(13), the mathematical expectation can be obtained as

$$\begin{cases} \mathbf{A}_j^{(i+1)} = \left(\hat{\mathbf{x}}_{j|k}^{(i+1)} - \mathbf{F}_j \hat{\mathbf{x}}_{j-1|k}^{(i+1)}\right)\left(\hat{\mathbf{x}}_{j|k}^{(i+1)} - \mathbf{F}_j \hat{\mathbf{x}}_{j-1|k}^{(i+1)}\right)^{\mathrm{T}} \\ + \mathbf{F}_j \mathbf{P}_{j-1|k}^{(i+1)} \mathbf{F}_j^{\mathrm{T}} - \left[\mathbf{P}_{j-1,j|k}^{(i+1)}\right]^{\mathrm{T}} \mathbf{F}_j^{\mathrm{T}} - \mathbf{F}_j \mathbf{P}_{j-1,j|k}^{(i+1)} + \mathbf{P}_{j|k}^{(i+1)} \\ \mathbf{B}_j^{(i+1)} = \mathbf{H}_j \mathbf{P}_{j|k}^{(i+1)} \mathbf{H}_j^{\mathrm{T}} + \left(\mathbf{z}_j - \mathbf{H}_j \hat{\mathbf{x}}_{j|k}^{(i+1)}\right)\left(\mathbf{z}_j - \mathbf{H}_j \hat{\mathbf{x}}_{j|k}^{(i+1)}\right)^{\mathrm{T}} \\ \mathrm{E}^{(i)}\left[\mathbf{Q}_k^{-1}\right] = \hat{y}_{k|k}^{(i)}\left\{\hat{\mathbf{Y}}_{k|k}^{(i)}\right\}^{-1}, \mathrm{E}^{(i)}\left[\mathbf{R}_k^{-1}\right] = \hat{u}_{k|k}^{(i)}\left\{\hat{\mathbf{U}}_{k|k}^{(i)}\right\}^{-1} \\ \mathrm{E}^{(i)}[\xi_j] = \hat{a}_{j|k}^{(i)} / \hat{b}_{j|k}^{(i)}, \mathrm{E}^{(i)}[\lambda_j] = \hat{c}_{j|k}^{(i)} / \hat{d}_{j|k}^{(i)} \end{cases} \tag{14}$$

where $\hat{\mathbf{x}}_{j|k}^{(i+1)}$ and $\mathbf{P}_{j|k}^{(i+1)}$ represent the smoothing estimate and the corresponding error covariance matrix at time j in the $(i+1)$th iteration, and $\mathbf{P}_{j-1,j|k}^{(i+1)}$ represents the smoothing errors cross-covariance matrix at times $j-1$ and j calculated as [29]

$$\mathbf{P}_{j-1,j|k}^{(i+1)} = \mathbf{G}_{j-1}^{(i+1)} \mathbf{P}_{j|k}^{(i+1)} \tag{15}$$

where $\mathbf{G}_{j-1}^{(i+1)}$ is the smoothing gain.

C. Outlier Detection Method Based on Cloud Model

The cloud model is a qualitative-quantitative transformation model [34]. It serves as an uncertainty transformation model that bridges a qualitative concept, represented by linguistic values, with its quantitative representation grounded in statistical and fuzzy mathematics. This model primarily captures the fuzziness and randomness inherent in concepts within the objective world or human knowledge, integrating both aspects to establish a mapping between qualitative and quantitative dimensions. The cloud model is characterized by three numerical features: expected value (Ex), entropy (En), and super entropy (He). The reverse cloud generator within the model can determine these three data features by inputting a specified number of cloud droplets. In practical positioning applications, model uncertainty can lead to state outliers, while adverse environmental conditions may result in measurement outliers, both of which significantly affect the final accuracy of positioning algorithms.

Consequently, relying solely on Kalman filtering methods proves inadequate for addressing the challenges posed by measurable state outliers. Considering that the cloud model can simultaneously address the coupling problem of "whether the state is abnormal" (fuzziness) and "whether the fluctuation is random" (randomness), it effectively reduces the misjudgment rate associated with traditional outlier detection methods. This article employs multiple positioning results to generate cloud droplets, introduces the principle of a reverse cloud generator to create three digital features, and subsequently calculates the certainty of each cloud droplet to assess the probability of outlier occurrences. This provides a foundation for establishing the parameters of the noise covariance matrix in the subsequent filtering process, thereby addressing the interference issues caused by state outliers and measurement outliers. The specific process is outlined as follows.

1) Calculate based on the recognition results of n sampled data at each moment The sample mean of this set of data is shown as follows:

$$\overline{\hat{\mathbf{F}}} = \frac{1}{n} \sum_{i=1}^{n} \hat{\mathbf{F}}_i \tag{16}$$

among them, $\mathbf{F}$ represents the sampled data, which can be multidimensional data, $\mathbf{F}_i = [x, y, v_x, w_z, w_z^{'}]$. Where x and y respectively represent the horizontal and vertical coordinates of the robot; $v_x, w_z, w_z^{'}$ Representing the linear velocity in the x direction, the angle of rotation around the z axis, and the angular velocity. n represents the length of

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the data segment used. First order sample absolute center distance

$$\frac{1}{n}\sum_{i=1}^{n}\left|\hat{\mathbf{F}}_i-\overline{\hat{\mathbf{F}}}\right| \tag{17}$$

sample variance

$$\mathbf{S}^2=\frac{1}{n-1}\sum_{i=1}^{n}\left(\hat{\mathbf{F}}_i-\overline{\hat{\mathbf{F}}}\right)^2. \tag{18}$$

2) Combining the theory of cloud models to calculate expectations, entropy, and super entropy

$$\hat{\mathbf{E}}_x=\hat{\mathbf{F}} \tag{19}$$

$$\hat{\mathbf{E}}_n=\sqrt{\frac{\pi}{2}}\times\frac{1}{n}\sum_{i=1}^{n}\left|\hat{\mathbf{F}}_i-\hat{\mathbf{E}}_x\right| \tag{20}$$

$$\hat{\mathbf{H}}_e=\sqrt{\left|\mathbf{S}^2-\hat{\mathbf{E}}_n^2\right|}. \tag{21}$$

3) Calculate the certainty of data using three numerical features

$$\mathbf{M}_i=\exp\left[-\left(\hat{\mathbf{F}}_i-\hat{\mathbf{E}}_x\right)^2/2\left(\hat{\mathbf{E}}_n\right)^2\right] \tag{22}$$

in the equation, $\hat{\mathbf{F}}_i$ represents the recognition result of the i-th sampled data at a certain moment in time.

4) Setting the parameters of the noise covariance matrix using certainty.

According to the properties of the cloud model, the closer the expected value is to the cloud, the higher the degree of certainty associated with that point. The expected value serves as the most representative qualitative concept of the point within the domain. Therefore, in this article, we calculate the degree of certainty, denoted as $\mathbf{M}_i$, corresponding to the recognition result of each sampled data point at each time interval. At this stage, we obtain n values of $\mathbf{M}_i$, and the recognition value corresponding to the maximum $\mathbf{M}_i$ is selected as the closest estimate of the true robot position at that time. The same process is applied for subsequent time intervals. Ultimately, determining the degree of deviation between the certainty of the current positioning data and the maximum certainty serves as the basis for establishing the noise covariance matrix. The judgment equation is as follows:

$$\mathbf{Q}_k=\begin{cases}\mathbf{Q}_0 & \text{if } \mathbf{M1}_{\max}-\mathbf{M1}_n<\mathbf{C_1}\\ 100\times\mathbf{Q}_0 & \text{else}\end{cases} \tag{23}$$

$$\mathbf{R}_k=\begin{cases}\mathbf{R}_0 & \text{if } \mathbf{M2}_{\max}-\mathbf{M2}_n<\mathbf{C_2}\\ 100\times\mathbf{R}_0 & \text{else}\end{cases} \tag{24}$$

among them, $\mathbf{R}_0$ and $\mathbf{R}_k$ are the covariance matrices of measurement noise, $\mathbf{Q}_0$ and $\mathbf{Q}_k$ are the covariance matrices of process noise. $\mathbf{M1}_{\max}$ and $\mathbf{M2}_{\max}$ respectively represent the maximum certainty of data used to determine and measure abnormal values in a certain period of time, while $\mathbf{M1}_n$ and $\mathbf{M2}_n$ respectively represent the certainty of data at the current time. The final parameters are determined based on the state values and measurement values at time n. $\mathbf{C_1}$ and $\mathbf{C_2}$ are numerical values that fall within a specified range.

Remark 1: It is important to note that the dimension of $\mathbf{M1}_{\max}$ corresponds to the state data used to identify outliers. For instance, the experimental section of this article estimates only the two-dimensional coordinates, x and y, of the robot, making $\mathbf{M1}_{\max}$ a two-dimensional vector. Additionally, we assess the maximum range of data fluctuations over a specified period based on the robot's motion pattern, which ultimately allows us to determine the value of $\mathbf{C}_1$. Similarly, relevant information regarding $\mathbf{M2}_{\max}$ can be inferred.

Remark 2: The cloud model is a cognitive framework that illustrates the relationship between randomness and fuzziness in a broad context. The likelihood of cloud droplets forming close to expected values is relatively high, signifying a strong degree of certainty and stability in representing a qualitative concept. Conversely, the probability of cloud droplets appearing far from expectations is relatively low, indicating a diminished and unstable certainty in representing that qualitative concept. In essence, the cloud model establishes a mapping that captures the interplay between randomness and fuzziness. The randomness of certainty reflects the variability and stability tendencies of this correlation. During the actual data sampling process, sensors return data at a specific frequency. We assume that, over a short period, the data will exhibit a distribution resembling a Gaussian distribution due to the influence of noise. In the absence of outliers, we calculate the expected value of the data within this time window and determine its corresponding degree of certainty. We continuously update the data within a sliding window and assess the difference between the certainty of the data at the end of the serial port and the expected value. If this difference is substantial, it suggests a high rate of change in the data, increasing the likelihood of outliers. Based on these evaluation results, this article proposes adjustments to the noise covariance matrix during the filtering process.

Remark 3: Unlike using the data features of the cloud model as a basis for updating the noise covariance matrix values at every moment, this article tests a cloud model-based method to dynamically adjust the noise covariance matrices Q and R prior to the algorithm's initiation. The results are illustrated in Fig. 3. In cases of significant initial value errors, this method enables the algorithm to converge to the true values in a remarkably short time. To facilitate the dynamic adjustment of noise covariance values during the positioning process, this article continues to employ the cloud model approach throughout the deployment process on the actual machine.

The implementation pseudocode of the proposed filter is shown in Algorithm 1, where ϵ represents the iteration threshold and N_m represents the maximum number of iterations. This article calculates the certainty of each data point (cloud droplet) before running the Kalman filter, in order to determine the likelihood of the data point being an outlier and adjust $\mathbf{R}_k$ and $\mathbf{Q}_k$.

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Algorithm 1: Pseudo Codes of the Proposed CM-SWRKF.

Inputs: $\{\hat{x}_{j|j}, \mathbf{P}_{j|j}, \hat{y}_{j|j}, \hat{Y}_{j|j}, \hat{u}_{j|j}, \hat{U}_{j|j}, \mathbf{z}_{j+1} \mid j \in [k-L, k-1]\}$ $\{\hat{\xi}_{j|k-1}, \hat{\lambda}_{j|k-1} \mid j \in [k-L+1, k-1]\}$ $\mathbf{F}_k, \mathbf{H}_k$, $\omega, \nu, \rho, L, m, n, \epsilon, \delta, N_m R_0, Q_0$

Initialization:

1. $\mathrm{E}^{(0)}[\mathbf{Q}_k^{-1}] = \hat{y}_{k-1|k-1}(\hat{\mathbf{Y}}_{k-1|k-1})^{-1}$
2. $\mathrm{E}^{(0)}[\mathbf{R}_k^{-1}] = \hat{u}_{k-1|k-1}(\hat{\mathbf{U}}_{k-1|k-1})^{-1}$

for $j = k-L+1$ **to** $k-1$ **do**

3. $\mathrm{E}^{(0)}[\xi_j] = \hat{\xi}_{j|k-1}$, $\mathrm{E}^{(0)}[\lambda_j] = \hat{\lambda}_{j|k-1}$

end for

4. Calculate $\hat{y}_{k|k-L}, \hat{\mathbf{Y}}_{k|k-L}, \hat{u}_{k|k-L}$, and $\hat{\mathbf{U}}_{k|k-L}$ using (3)

Iteration:

for $i = 0 : N_m - 1$ **do**

Update $q^{(i+1)}(\mathbf{x}_{k-L:k})$ given $q^{(i+1)}(\mathbf{Q}_k)$, $q^{(i+1)}(\mathbf{R}_k)$, $q^{(i+1)}(\xi_{k-L+1:k})$, and $q^{(i+1)}(\lambda_{k-L+1:k})$

5. $\hat{\mathbf{x}}_{k-L|k-L}^{(i+1)} = \hat{\mathbf{x}}_{k-L|k-L}$, $\hat{\mathbf{P}}_{k-L|k-L}^{(i+1)} = \hat{\mathbf{P}}_{k-L|k-L}$

for $j = k-L+1 : k$ **do**

Run cloud model 6.

Calculate $\hat{\mathbf{E}}_x, \hat{\mathbf{E}}_n, \hat{\mathbf{H}}_e$

7. $\mathbf{M_i} = \exp\left[-\left(\hat{\mathbf{F}}_i - \hat{\mathbf{E}}_x\right)^2 / 2\left(\hat{\mathbf{E}}_n\right)^2\right]$

8. Calculate $\mathbf{M_i}$ and set $\mathbf{Q}_k, \mathbf{R}_k$

Run the conventional KF with modified SNCM $\frac{\mathrm{E}^{(i)}[\mathbf{Q}_k^{-1}]^{-1}}{\mathrm{E}^{(i)}[\xi_j]}$ and MNCM $\frac{\mathrm{E}^{(i)}[\mathbf{R}_k^{-1}]^{-1}}{\mathrm{E}^{(i)}[\lambda_j]}$

9. $\hat{\mathbf{x}}_{j|j-1}^{(i+1)} = \mathbf{F}_j \hat{\mathbf{x}}_{j-1|j-1}^{(i+1)}$

10. $\mathbf{P}_{j|j-1}^{(i+1)} = \mathbf{F}_j \mathbf{P}_{j-1|j-1}^{(i+1)} \mathbf{F}_j^{\mathrm{T}} + \frac{\mathrm{E}^{(i)}[\mathbf{Q}_k^{-1}]^{-1}}{\mathrm{E}^{(i)}[\xi_j]}$

11. $\mathbf{K}_j^{(i+1)} = \mathbf{P}_{j|j-1}^{(i+1)} \mathbf{H}_j^{\mathrm{T}} (\mathbf{H}_j \mathbf{P}_{j|j-1}^{(i+1)} \mathbf{H}_j^{\mathrm{T}} + \frac{\mathrm{E}^{(i)}[\mathbf{R}_k^{-1}]^{-1}}{\mathrm{E}^{(i)}[\lambda_j]})^{-1}$

12. $\hat{\mathbf{x}}_{j|j}^{(i+1)} = \hat{\mathbf{x}}_{j|j-1}^{(i+1)} + \mathbf{K}_j^{(i+1)}(\mathbf{z}_j - \mathbf{H}_j \hat{\mathbf{x}}_{j|j-1}^{(i+1)})$

13. $\mathbf{P}_{j|j}^{(i+1)} = (\mathbf{I}_n - \mathbf{K}_j^{(i+1)} \mathbf{H}_j) \mathbf{P}_{j|j-1}^{(i+1)}$

end for

for $j = k : -1 : k-L+1$ **do**

Run the conventional KS with modified SNCM $\frac{\mathrm{E}^{(i)}[\mathbf{Q}_k^{-1}]^{-1}}{\mathrm{E}^{(i)}[\xi_j]}$

14. $\mathbf{G}_{j-1}^{(i+1)} = \mathbf{P}_{j-1|j-1}^{(i+1)} \mathbf{F}_j^{\mathrm{T}} [\mathbf{P}_{j|j-1}^{(i+1)}]^{-1}$

15. $\hat{\mathbf{x}}_{j-1|k}^{(i+1)} = \hat{\mathbf{x}}_{j-1|j-1}^{(i+1)} + \mathbf{G}_{j-1}^{(i+1)}(\hat{\mathbf{x}}_{j|k}^{(i+1)} - \hat{\mathbf{x}}_{j|j-1}^{(i+1)})$

16. $\mathbf{P}_{j-1|k}^{(i+1)} = \mathbf{P}_{j-1|j-1}^{(i+1)} + \mathbf{G}_{j-1}^{(i+1)}(\mathbf{P}_{j|k}^{(i+1)} - \mathbf{P}_{j|j-1}^{(i+1)})[\mathbf{G}_{j-1}^{(i+1)}]^{\mathrm{T}}$

end for

Update $q^{(i+1)}(\xi_{k-L+1:k})$ and $q^{(i+1)}(\lambda_{k-L+1:k})$ given $q^{(i+1)}(\mathbf{x}_{k-L:k})$

17. Calculate $\mathbf{A}_j^{(i+1)}$ and $\mathbf{B}_j^{(i+1)}$ using (14)

18 Calculate $\hat{a}_{j|k}^{(i+1)}$, $\hat{b}_{j|k}^{(i+1)}$, $\hat{c}_{j|k}^{(i+1)}$, and $\hat{d}_{j|k}^{(i+1)}$ using (13)

19. Calculate $\mathrm{E}^{(i+1)}[\xi_j]$ and $\mathrm{E}^{(i+1)}[\lambda_j]$ using (14)

20. Calculate $\hat{y}_{k|k}^{(i+1)}$, $\hat{\mathbf{Y}}_{k|k}^{(i+1)}$, $\hat{u}_{k|k}^{(i+1)}$, and $\hat{\mathbf{U}}_{k|k}^{(i+1)}$ using (10)

21. Calculate $\mathrm{E}^{(i+1)}[\mathbf{Q}_k^{-1}]$ and $\mathrm{E}^{(i+1)}[\mathbf{R}_k^{-1}]$ using (14)

22.

if $\left\|\hat{\mathbf{x}}_{k|k}^{(i+1)} - \hat{\mathbf{x}}_{k|k}^{(i)}\right\| / \left\|\hat{\mathbf{x}}_{k|k}^{(i)}\right\| \leq \epsilon$ **then**

terminate iterations

end if

end for

Data saving:

for $j = k-L+2 : k$ **do**

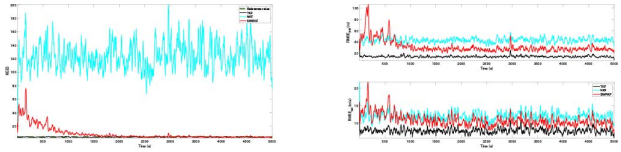
23. $\hat{\xi}_{j|k} = \mathrm{E}^{(i+1)}[\xi_j]$, $\hat{\lambda}_{j|k} = \mathrm{E}^{(i+1)}[\lambda_j]$

end for

24. $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k}^{(i+1)}$, $\mathbf{P}_{k|k} = \mathbf{P}_{k|k}^{(i+1)}$, $\hat{y}_{k|k} = \hat{y}_{k|k}^{(i+1)}$, $\hat{\mathbf{Y}}_{k|k} = \hat{\mathbf{Y}}_{k|k}^{(i+1)}$

25. $\hat{u}_{k|k} = \hat{u}_{k|k}^{(i+1)}$, $\hat{\mathbf{U}}_{k|k} = \hat{\mathbf{U}}_{k|k}^{(i+1)}$

Outputs: $\{\hat{\mathbf{x}}_{j|j}, \mathbf{P}_{j|j}, \hat{y}_{j|j}, \hat{\mathbf{Y}}_{j|j}, \hat{u}_{j|j}, \hat{\mathbf{U}}_{j|j} \mid j \in [k-L+1, k]\}$ $\{\hat{\xi}_{j|k}, \hat{\lambda}_{j|k} \mid j \in [k-L+2, k]\}$



(a) (b)

Fig. 2. Comparison of estimation results for algorithms TKF, NKF, and SWRKF. (a) NEES. (b) RMSE.

III. Experimentations

A. Simulation Study

This section uses KF with true noise covariance matrix (TKF), KF with nominal noise covariance matrix (NKF), and KF with sliding window (SWRKF) based on Student's t-noise model as simulation references. Compare the performance of the KF algorithm with sliding window (CM-SWRKF) based on Student's t-noise model and cloud model anomaly detection in this article.

This section demonstrates the advantages of the proposed CM-SWRKF by simulating the tracking of manipulated targets. In target tracking simulation, we define the state vector as $\mathbf{x}_k = [x_k, y_k, \dot{x}_k, \dot{y}_k]^{\mathrm{T}}$, where x_k and y_k represent the horizontal and vertical coordinates of the target, respectively, and $\dot{x}_k$ and $\dot{y}_k$, represent the corresponding velocities. The kinematic and measurement models are described as

$$\begin{cases} \mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{w}_k \\ \mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k \end{cases} \tag{25}$$

in which the state transition and measurement matrices are

$$\mathbf{F}_k = \begin{bmatrix} \mathbf{I}_2 & \Delta t \mathbf{I}_2 \\ \mathbf{0} & \mathbf{I}_2 \end{bmatrix}, \quad \mathbf{H}_k = \begin{bmatrix} \mathbf{I}_2 & \mathbf{0} \end{bmatrix} \tag{26}$$

and the outlier-contaminated noise is generated as described as (23) and (24). The root mean square error (RMSE) and normalized estimation error squared (NEES) are utilized to assess estimation accuracy and consistency. The RMSE and NEES values for different algorithms, computed based on 1,000 Monte Carlo simulations, are presented in Fig. 2.

The comparison chart of NEES estimation of TKF, NKF, and SWRKF algorithms with the maximum accuracy values of (a = 0.1, 0.2, 0.3, 0.4) for the current data accuracy and the data within the time window is shown in Figs. 3 and 4. At this point, the probability of the occurrence of outliers is set to p = 0.3.

Comparing Figs. 2–4, it is evident that the improved algorithm proposed in this article demonstrates superior positioning accuracy and faster convergence speed. For instance, regarding convergence speed, the SWRKF algorithm gradually converges after 1500 time steps, whereas the CM-SWRKF algorithm converges in approximately 80 time steps, thereby validating the effectiveness of the proposed method. In terms of estimation accuracy, the algorithm presented in this article slightly outperforms other algorithms based on the RMSE metric, with the relevant statistical data displayed in Tables I and II.

Notably, the algorithm maintains high real-time performance while ensuring convergence speed. In the MATLAB simulation, it was observed that compared with SWRKF, the CM-SWRKF

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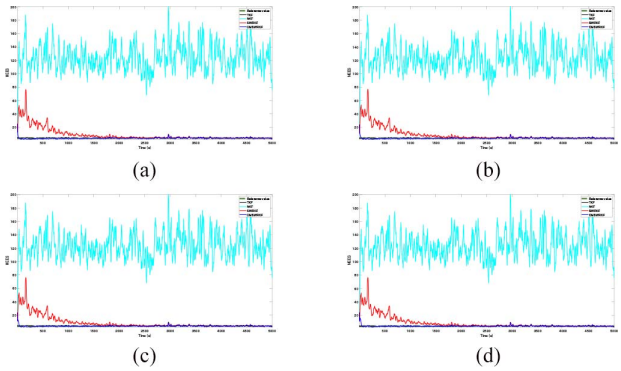


Fig. 3. Comparison of NEES estimation results for different algorithms. (a) p = 0.3 and a = 0.1. (b) p = 0.3 and a = 0.2. (c) p = 0.3 and a = 0.3. (d) p = 0.3 and a = 0.4.

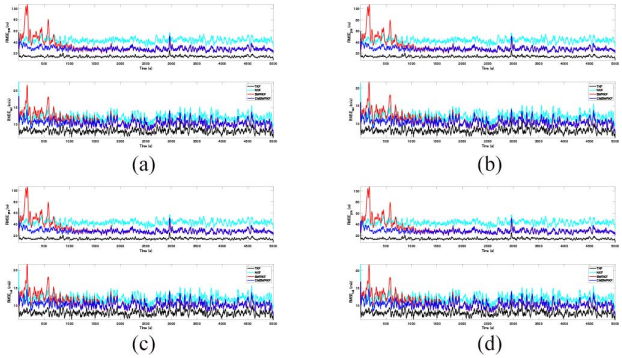


Fig. 4. Comparison of RMSE estimation results for different algorithms. (a) p = 0.3 and a = 0.1. (b) p = 0.3 and a = 0.2. (c) p = 0.3 and a = 0.3. (d) p = 0.3 and a = 0.4.

TABLE I
COMPARISON OF POSITION AND VELOCITY ERRORS FOR DIFFERENT ALGORITHMS UNDER DIFFERENT VALUES OF a WITH $p = 0.1$

Algorithms	Unit(m or m/s)	0.1	0.2	0.3	0.4	0.5
TKF	PE	9.77	9.77	9.77	9.77	9.77
	SE	4.59	4.59	4.59	4.59	4.59
NKF	PE	25.23	25.23	25.23	25.23	25.23
	SE	7.02	7.02	7.02	7.02	7.02
SWRKF	PE	13.17	13.17	13.17	13.17	13.17
	SE	5.60	5.60	5.60	5.60	5.60
CMSWRKF	PE	13.063	13.058	13.067	13.062	13.060
	SE	5.555	5.555	5.557	5.555	5.556

Note: PE: Position Error. SE: Speed Error.

algorithm requires only 1.4 seconds more after running 10 iterations of 5000 time steps each. Furthermore, in the real-world experiment, this article deploys the algorithm on a wheeled robot, and the experimental process consistently maintains a filtering output frequency of 50 Hz.

From Tables I and II, it is evident that as the probability of outliers increases, the filtering method that incorporates the cloud model demonstrates improved state estimation accuracy. Notably, TKF, NKF, and SWRKF do not utilize cloud models; therefore, variations in the probability of outliers do not affect their estimation results.

TABLE II
COMPARISON OF POSITION AND VELOCITY ERRORS FOR DIFFERENT ALGORITHMS UNDER DIFFERENT VALUES OF a WITH $p = 0.3$

Algorithms	Unit(m or m/s)	0.1	0.2	0.3	0.4	0.5
TKF	PE	14.18	14.18	14.18	14.18	14.18
	SE	7.90	7.90	7.90	7.90	7.90
NKF	PE	42.89	42.89	42.89	42.89	42.89
	SE	11.97	11.97	11.97	11.97	11.97
SWRKF	PE	30.48	30.48	30.48	30.48	30.48
	SE	10.75	10.75	10.75	10.75	10.75
CMSWRKF	PE	27.95	27.91	27.89	27.82	27.66
	SE	10.31	10.31	10.30	10.29	10.29

Note: PE: Position Error. SE: Speed Error.

B. Real Machine Experiment

Simulation experiments have confirmed the effectiveness of the algorithm. To further assess its practicality, this article deploys the algorithm on a real machine for position estimation experiments. This article primarily presents two experiments. Experiment 1 has a 10% probability of outlier occurrence, while Experiment 2 has a 30% probability. To simulate the occurrence of outliers, this article generates outliers by randomly magnifying the angle and speed information of the sensor data by a factor of 2. If the probability of an outlier is 0.1, then one outlier will occur for every ten data points. In a harsh pipeline environment-simulated using a softer texture pipeline to replicate the conditions of underground pipelines-issues such as tilting, pitching, and slipping may arise during the positioning process. Abnormal velocity and angular velocity data are generated through random sampling. Subsequently, the algorithm proposed in this article is compared with existing algorithms for experimental evaluation.

This study presents a pipeline scenario with a diameter of 300 mm. Given the issue of wheel slippage in wet and slippery pipelines, a soft and collapsible pipeline is selected to simulate the urban underground drainage environment, which may lead to some degree of wheel sliding. During the experiment, a four-wheeled vehicle operated as a pipeline clearing robot and traversed the pipeline, covering a distance of over 120 m. The motion was recorded using a wheel encoder (500 ppr, 20 Hz) and inertial measurement unit (IMU) data sampled at 50 Hz (the resolution of the accelerometer is 0.0005 g/LSB, and the resolution of the gyroscope is 0.061 (°/s)/(LSB)), while the robot's position was determined through a motion capture system, which had an error margin of 0.63 mm. This provided accurate values for subsequent calculations of positioning errors. The true values of the robot's motion trajectory in this experiment are illustrated in Fig. 5, where colors represent time series, with lighter colors indicating later time points.

The experiments were conducted on a 64-bit operating system with four cores and 16 GB of memory, featuring a CPU processing frequency of 2 GHz, running Ubuntu 20.04, and utilizing the robot operating system version noetic. The experimental setup and robot platform used are illustrated in Fig. 6.

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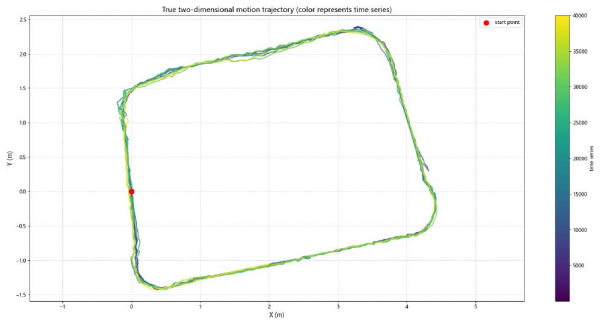


Fig. 5. Real trajectory.



(a) (b)

Fig. 6. Experimental platform: motion capture system, pipeline simulation scene, and four-wheel robot. (a) Experimental site. (b) Four wheeled robot.

This article utilizes recorded sensor data as input for measurements and integrates the EKF algorithm with the noise model proposed herein to conduct robot localization experiments. To address computational power constraints in practical applications, this article focuses solely on reconstructing the measurement noise model; specifically, it employs the method introduced in this study to update the measurement noise covariance matrix during the position estimation process.

We define the state vector as $\boldsymbol{X}_k = [x, y, v_x, w_z, w_z^{'}]$, where $v_x, w_z, w_z^{'}$ represent the linear velocity in the x direction, the angle of rotation around the z axis, and the angular velocity. The kinematic and measurement models are described as

$$\begin{cases} \boldsymbol{X}_k = \boldsymbol{F}\left(\boldsymbol{X}_{k-1}, \boldsymbol{u}_k\right) + \boldsymbol{B_{k-1}} \boldsymbol{w}_k \\ \boldsymbol{Y}_k = \boldsymbol{G}\left(\boldsymbol{X}_k\right) + \boldsymbol{C}_k \boldsymbol{n}_k \end{cases} \tag{27}$$

in which the state transition and measurement matrices are

$$\boldsymbol{F} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \cdots & \frac{\partial f}{\partial x_5} \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_5} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_5}{\partial x_1} & \cdots & \frac{\partial f_5}{\partial x_5} \end{bmatrix} \tag{28}$$

where $[x_1, x_2, x_3, x_4, x_5]$ represent the five elements of $\boldsymbol{X}$ respectively. f represents a nonlinear functional relationship, determined based on the vehicle motion model. This article uses a constant turning rate and speed model. The specific derivation process will not be explained in detail

$$\boldsymbol{G} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}. \tag{29}$$

For the noise covariance matrix in the filtering process, after adjusting the EKF algorithm to a relatively ideal performance,

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS

Indicators	Algorithms ($p = 0.1$)	Max	Mean	Median	Min	RMSE	SSE	Std.
ape(unit/m)	EKF	2.934	1.546	1.496	0.423	0.423	31 135.499	0.570
	AEKF	2.951	2.045	2.027	1.341	2.081	49 859.236	0.382
	SWREKF	2.801	1.027	0.923	0.011	1.158	15 019.730	0.536
	CM-SWREKF	**2.327**	**0.860**	**0.800**	0.032	0.970	**10 741.098**	**0.449**
rpe(unit/m)	EKF	0.161	0.008	0.005	0.000	0.014	2.127	0.011
	AEKF	0.131	0.010	0.007	0.000	0.016	2.950	0.012
	SWREKF	0.155	0.009	0.006	0.000	0.014	2.279	0.011
	CM-SWREKF	0.163	0.009	0.006	0.000	0.014	2.288	0.011

Note: Bold entries indicate the algorithm performance.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT ALGORITHMS

Indicators	Algorithms ($p = 0.3$)	Max	Mean	Median	Min	RMSE	SSE	Std.
ape(unit/m)	EKF	3.183	1.926	1.839	0.918	2.003	45 246.426	0.550
	AEKF	3.010	2.047	2.021	1.333	2.082	49 141.329	0.377
	SWREKF	2.596	0.983	0.953	1.333	2.089	13 521.881	0.469
	CM-SWREKF	**2.216**	**0.935**	**0.810**	**0.021**	**1.052**	**12 555.898**	0.482
rpe(unit/m)	EKF	0.136	0.009	0.006	0.000	0.015	2.509	0.012
	AEKF	0.160	0.010	0.007	0.000	0.016	2.915	0.012
	SWREKF	0.130	0.009	0.006	0.000	0.014	2.298	0.011
	CM-SWREKF	0.158	0.009	0.006	0.000	0.014	2.318	0.011

Note: Bold entries indicate the algorithm performance.

taking the parameters of this algorithm as the benchmark, this article sets $\boldsymbol{Q}_0$ and $\boldsymbol{R}_0$ as the initial values

$$\boldsymbol{R_0} = \begin{bmatrix} 0.07 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.07 \end{bmatrix} \tag{30}$$

$$\boldsymbol{Q_0} = \begin{bmatrix} 0.03 & 0 & 0 & 0 & 0 \\ 0 & 0.03 & 0 & 0 & 0 \\ 0 & 0 & 0.03 & 0 & 0 \\ 0 & 0 & 0 & 0.03 & 0 \\ 0 & 0 & 0 & 0 & 0.03 \end{bmatrix}. \tag{31}$$

In the actual deployment process, this article utilized the standard EKF algorithm as a benchmark to compare the adaptive extended Kalman filter (AEKF), the EKF with a sliding window based on Student's t-noise model (SWREKF), and the CM-SWREKF algorithm that employs cloud models. The experimental data are presented in Tables III and IV. When comparing the performance of algorithms, we resampled the data collected by the motion capture system to 50 Hz after aligning the ROS timestamps, saved the data in txt data format, and used the evo trajectory evaluation tool for performance assessment.

In engineering applications, the emphasis is on achieving global consistency of trajectories; therefore, this article primarily discusses Absolute Pose Error (APE). As shown in Tables III and IV, it can be seen that as the probability of outliers increases, the method proposed in this article demonstrates superior performance in maintaining global consistency of trajectories, with both maximum and average errors being smaller than those of other algorithms.

On one hand, when the probability of outliers is 0.1, the improved algorithm presented in this article reduces the maximum position error by 20.69% and the average position error by 44.37% compared with the traditional EKF algorithm. On the other hand, when the probability of outliers increases to 0.3, the maximum position error is reduced by 30.38%, the average position error is reduced by 51.45%. Additionally, when compared

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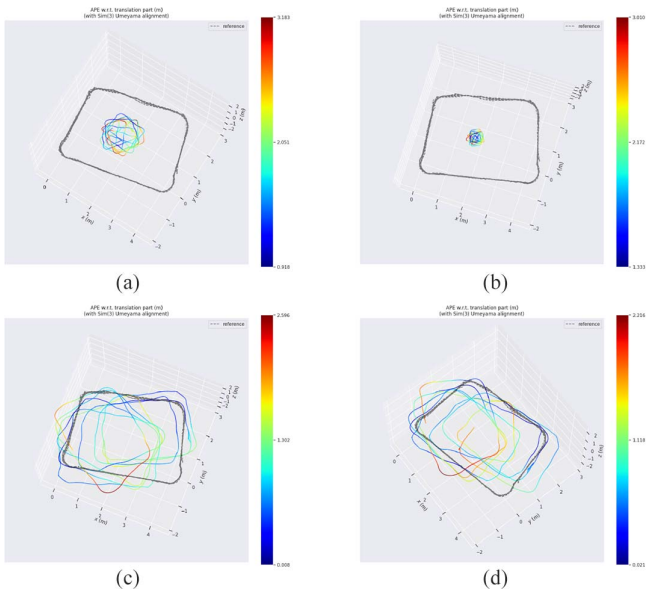


Fig. 7. Positioning performance of different algorithms under the condition of $p = 0.3$. (a) Algorithm EKF. (b) Algorithm AEKF. (c) Algorithm SWREKF. (d) Algorithm CM-SWREKF.

with the algorithms such as AEKF and SWREKF, the error is also reduced to varying degrees.

In addition, upon reviewing the data, it is evident that the second most effective algorithm for positioning accuracy is the SWREKF, which is implemented as described in reference [29] and does not utilize cloud models to adjust the parameters of the covariance matrix. When $p = 0.1$, compared with the CM-SWREKF algorithm, its maximum positioning error and average positioning error are 16.9% and 16.28% higher, respectively. Similarly, when $p = 0.3$, the maximum and average positioning errors are 14.64% and 4.9% higher, respectively. This indicates that the algorithm proposed in this article demonstrates superior estimation accuracy in the presence of outliers.

To more intuitively demonstrate the effectiveness of algorithm improvements, this article will plot and list the trajectories of various algorithms when confronted with different outlier probabilities, as illustrated in Fig. 7.

On one hand, as illustrated in Fig. 7, the filtering algorithm utilizing the Student's t-noise model can effectively maintain the trajectory shape, even in the presence of outliers within the measurement data. In contrast, traditional EKF and AEKF algorithms struggle to accurately estimate position as the likelihood of outliers increases, leading to unreliable trajectory representations. On the other hand, the presence of pitch, roll, and slip in the robot's motion posture within the simulated pipeline environment complicates the use of visual sensors and other methods for loop detection in narrow diameters, making it challenging to correct accumulated angular errors. Consequently, the effectiveness of relying solely on data from wheel encoders and IMUs for positioning remains inadequately estimated, even with enhanced algorithms. While Student's t-distribution noise models and cloud model theories demonstrate enhanced outlier robustness, implementation methods remain limited. Furthermore, global attitude coordinate transformation requires further research in engineering applications due to insufficient global information. The algorithm's performance in actual deployments is expected to closely align with real-machine experimental results.

IV. Conclusion

In summary, this article first introduces an early noise model based on the Student's t-distribution, employs the VB method to calculate the joint distribution probability, and integrates the inference results with Kalman filtering. Subsequently, a noise parameter adjustment strategy based on cloud models is proposed, offering a novel solution for mitigating the impact of outliers. The final experimental results demonstrate the algorithm's effectiveness. However, the available positioning solutions in urban underground small pipe environments are limited, making it challenging to achieve accurate position estimation using only wheel encoders and IMUs. Based on practical needs, future research directions will explore the integration of Long Short-Term Memory networks. By analyzing the implicit features of the data, loop detection for positioning can be achieved, thereby reducing cumulative errors in both angle and position. Ultimately, accurate position estimation is attained through the application of Kalman filtering theory.

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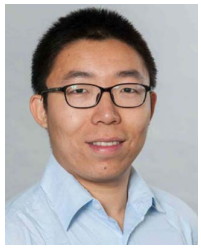
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